INCH-POUND

MIL-M-28787/252
AMENDMENT 2
28 April 1993
SUPERSEDING
AMENDMENT 1
16 August 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND-OR-INVERT, DIGITAL, KEY CODE AMH

This amendment forms a part of Military Specification MIL-M-28787/252(EC), dated 23 July 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through TT shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, and C where applicable, followed by the remaining tests in the sequence."

PAGES 4, 5, 6, 7, 8, AND 9

*TABLE I: Delete and substitute new table I as printed on pages 4, 5, 6, 7, 8, 9, 10, 11, and 12 of this amendment.

PAGE 10

*4.5.1, fifth line: Delete "B through M," and substitute "D through M,".

4.5.1, tenth line: Delete "100°C" and substitute "85°C".

*4.5.3, third line: Delete "B through PP" and substitute "D through PP".

*4.5.3, sixth line: Delete "B through TT" and substitute "D through TT".

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PAGE 11

6.4, Output drive current (V_{DD} = 5V): Delete "100°C" (two places) and substitute "85°C" (two places).

PAGE 20

FIGURE 6: Delete and substitute new figure 6 as printed on page 13 of this amendment.

PAGE 24

FIGURE 8: Delete and substitute new figure 8 as printed on page 14 of this amendment.

PAGE 26

FIGURE 9: Delete and substitute new figure 9 as printed on page 15 of this amendment.

PAGE 27

FIGURE 10: Delete and substitute new figure 10 as printed on page 16 of this amendment.

PAGE 29

FIGURE 11: Delete and substitute new figure 11 as printed on page 17 of this amendment.

PAGE 34

FIGURE 12: Delete and substitute new figure 12 as printed on page 18 of this amendment.

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*FIGURE 31, ITEM C01 box: Delete "C01" and substitute "C06".

*FIGURE 31, NOTE 3, first line: Delete "C01" and substitute "C06".

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NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Navy - SH
Air Force - 85

Review activities:
Army - AT
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Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0204)

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B30	Input clamp diode voltage, negative	IV	S	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C30	Input clamp diode voltage, positive	V	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D6	Output leakage current, high level, VDD = 15 V	VI	X	-0.10	0.10	-0.10	0.10	-1.45	1.45	-0.10	0.10	-2.4	2.4	μA
E1-E6	Output leakage current, low level, VDD = 15 V	VII	X	-0.10	0.10	-0.10	0.10	-1.45	1.45	-0.10	0.10	-2.4	2.4	μA
F1-F6	Static output voltage, high level, VDD = 5 V (IOH = 0)	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G6	Static output voltage, low level, VDD = 5 V (IOL = 0)	IX	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
H1-H6	Static output voltage, high level, VDD = 5 V	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J6	Static output voltage, low level, VDD = 5 V	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
K1-K6	Static output voltage, high level, VDD = 15 V	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L6	Static output voltage, low level, VDD = 15 V	XIII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
M1-M64	Functional test	XIV	X	<---------- See table XIV ---------->										
N1-N30	Input current, high level	XV	X	0	200	0	200	0	200	---	---	0	200	nA
P1-P30	Input current, low level	XVI	X	0	-200	0	-200	0	-200	---	---	0	-200	nA
Q1	Static power supply current, low level	XVII	X	0	12	0	12	0	175	0	12	0	291	µA
Q2	Static power supply current, high level	XVII	X	0	12	0	12	0	175	0	12	0	291	µA
R1-R6	Transition time (tTLH), CL = 50 pF, VDD = 5 V	XVIII	S	35	350	35	350	38	385	26	265	41	413	ns
R7-R12	Transition time (tTLH), CL = 150 pF, VDD = 5 V	XVIII	Q	83	830	83	830	91	915	63	630	97	978	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S6	Transition time (tTHL), CL = 50 pF, VDD = 5 V	XIX	S	17	175	17	175	19	195	13	135	20	208	ns
S7-S12	Transition time (tTHL), CL = 150 pF, VDD = 5 V	XIX	Q	65	655	65	655	72	725	50	500	81	819	ns
T1-T24	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XX	S	44	440	44	440	48	485	33	335	52	519	ns
T25-T30	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XX	S	27	270	27	270	30	300	20	205	32	319	ns
T31-T36	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XX	S	36	360	36	360	40	400	27	275	43	425	ns
T37-T60	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XX	Q	64	640	64	640	70	705	48	485	75	755	ns
T61-T66	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XX	Q	47	470	47	470	52	520	35	355	55	554	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T67-T72	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XX	Q	56	560	56	560	62	620	42	425	66	661	ns
U1-U24	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XXI	S	36	360	36	360	40	400	27	275	43	425	ns
U25-U30	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XXI	S	27	270	27	270	30	300	20	205	32	319	ns
U31-U36	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XXI	S	36	360	36	360	40	400	27	275	43	425	ns
U37-U60	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XXI	Q	56	560	56	560	62	620	42	425	66	661	ns
U61-U66	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XXI	Q	47	470	47	470	52	520	35	355	55	554	ns
U67-U72	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XXI	Q	56	560	56	560	62	620	42	425	66	661	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V1-V6	Disable delay time (tPZH), CL = 50 pF, VDD = 5 V	XXII	S	25	250	25	250	27	275	19	190	29	294	ns
V7-V12	Disable delay time (tPZH), CL = 150 pF, VDD = 5 V	XXII	Q	45	450	45	450	50	500	34	340	53	531	ns
W1-W6	Disable delay time (tPHZ), CL = 50 pF, VDD = 5 V	XXIII	S	25	250	25	250	26	265	22	220	27	274	ns
W7-W12	Disable delay time (tPHZ), CL = 150 pF, VDD = 5 V	XXIII	Q	45	450	45	450	48	485	37	370	51	510	ns
X1-X6	Disable delay time (tPZL), CL = 50 pF, VDD = 5 V	XXIV	S	13	135	13	135	15	150	10	100	16	159	ns
X7-X12	Disable delay time (tPZL), CL = 150 pF, VDD = 5 V	XXIV	Q	33	335	33	335	37	370	25	255	40	395	ns
Y1-Y6	Disable delay time (tPLZ), CL = 50 pF, VDD = 5 V	XXV	S	25	255	25	255	28	280	19	195	30	302	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y7-Y12	Disable delay time (tPLZ), CL = 150 pF, VDD = 5 V	XXV	Q	45	455	45	455	50	505	34	345	53	536	ns
Z1-Z6	Transition time (tTLH), CL = 50 pF, VDD = 10 V	XXVI	Q	15	150	15	150	16	165	11	115	17	178	ns
AA1-AA6	Transition time (tTHL), CL = 50 pF, VDD = 10 V	XXVII	Q	7	75	7	75	8	83	5	57	9	89	ns
BB1-BB24	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXVIII	Q	16	165	16	165	18	185	12	125	19	194	ns
BB25-BB30	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXVIII	Q	11	110	11	110	12	120	8	85	13	129	ns
BB31-BB36	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXVIII	Q	15	150	15	150	16	165	11	115	17	178	ns
CC1-CC24	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXIX	Q	13	135	13	135	15	150	10	105	16	159	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC25-CC30	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXIX	Q	9	90	9	90	9	99	6	68	10	106	ns
CC31-CC36	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXIX	Q	14	140	14	140	15	155	10	105	16	164	ns
DD1-DD6	Disable delay time (tPZH), CL = 50 pF, VDD = 10 V	XXX	Q	10	100	10	100	11	110	7	75	12	119	ns
EE1-EE6	Disable delay time (tPHZ), CL = 50 pF, VDD = 10 V	XXXI	Q	10	100	10	100	11	110	8	80	12	116	ns
FF1-FF6	Disable delay time (tPZL), CL = 50 pF, VDD = 10 V	XXXII	Q	10	100	10	100	11	110	8	80	12	116	ns
GG1-GG6	Disable delay time (tPLZ), CL = 50 pF, VDD = 10 V	XXXIII	Q	10	105	10	105	11	115	8	80	12	124	ns
HH1-HH6	Transition time (tTLH), CL = 50 pF, VDD = 15 V	XXXIV	Q	11	110	11	110	12	120	8	85	13	129	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
JJ1-JJ6	Transition time (tTHL), CL = 50 pF, VDD = 15 V	XXXV	Q	5	55	5	55	6	61	4	42	6	65	ns
KK1-KK24	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXXVI	Q	11	110	11	110	12	120	8	85	13	129	ns
KK25-KK30	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXXVI	Q	7	75	7	75	8	83	5	57	9	89	ns
KK31-KK36	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXXVI	Q	10	100	10	100	11	110	7	75	12	119	ns
LL1-LL24	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXXVII	Q	11	110	11	110	12	120	8	85	13	129	ns
LL25-LL30	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXXVII	Q	6	60	6	60	6	66	4	45	7	71	ns
LL31-LL36	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXXVII	Q	10	100	10	100	11	110	7	75	12	119	ns

- See footnotes at end of table.

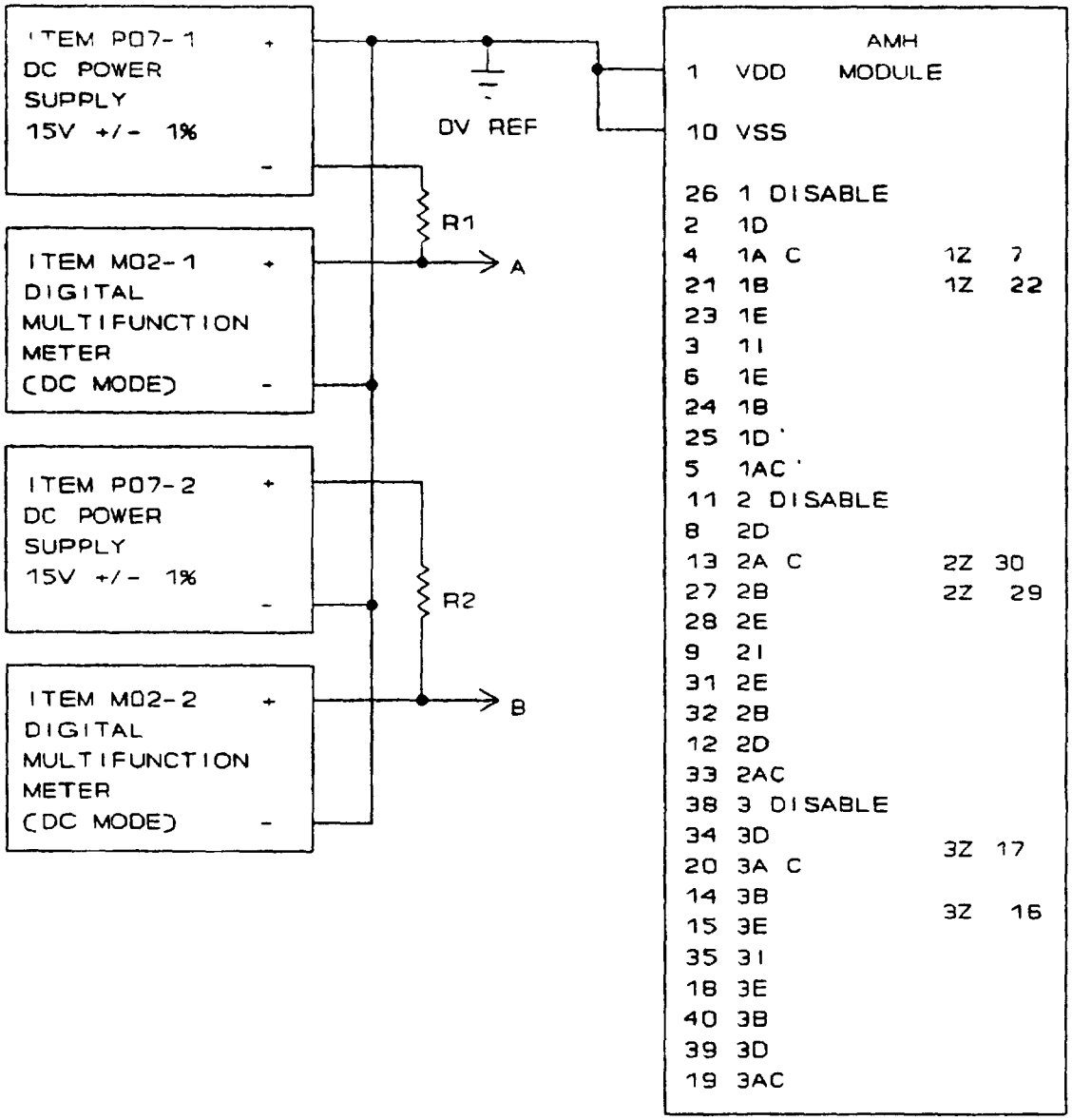
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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
MM1-MM6	Disable delay time (tPZH), CL = 50 pF, VDD = 15 V	XXXVIII	Q	8	80	8	80	8	86	6	65	9	92	ns
NN1-NN6	Disable delay time (tPHZ), CL = 50 pF, VDD = 15 V	XXXIX	Q	8	80	8	80	8	86	6	65	9	92	ns
PP1-PP6	Disable delay time (tPZL), CL = 50 pF, VDD = 15 V	XL	Q	8	80	8	80	8	86	6	65	9	92	ns
QQ1-QQ6	Disable delay time (tPLZ), CL = 50 pF, VDD = 15 V	XLI	Q	8	80	8	80	8	86	6	65	9	92	ns
RR	Power supply filter capacitance	XLII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
SS1-SS24	Input capacitance	XLIII	Q	2	15	2	15	2	15	2	15	2	15	pF
SS25-SS30	Input capacitance	XLIII	Q	4	25	4	25	4	25	4	25	4	25	pF
TT1-TT6	Output capacitance	XLIV	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

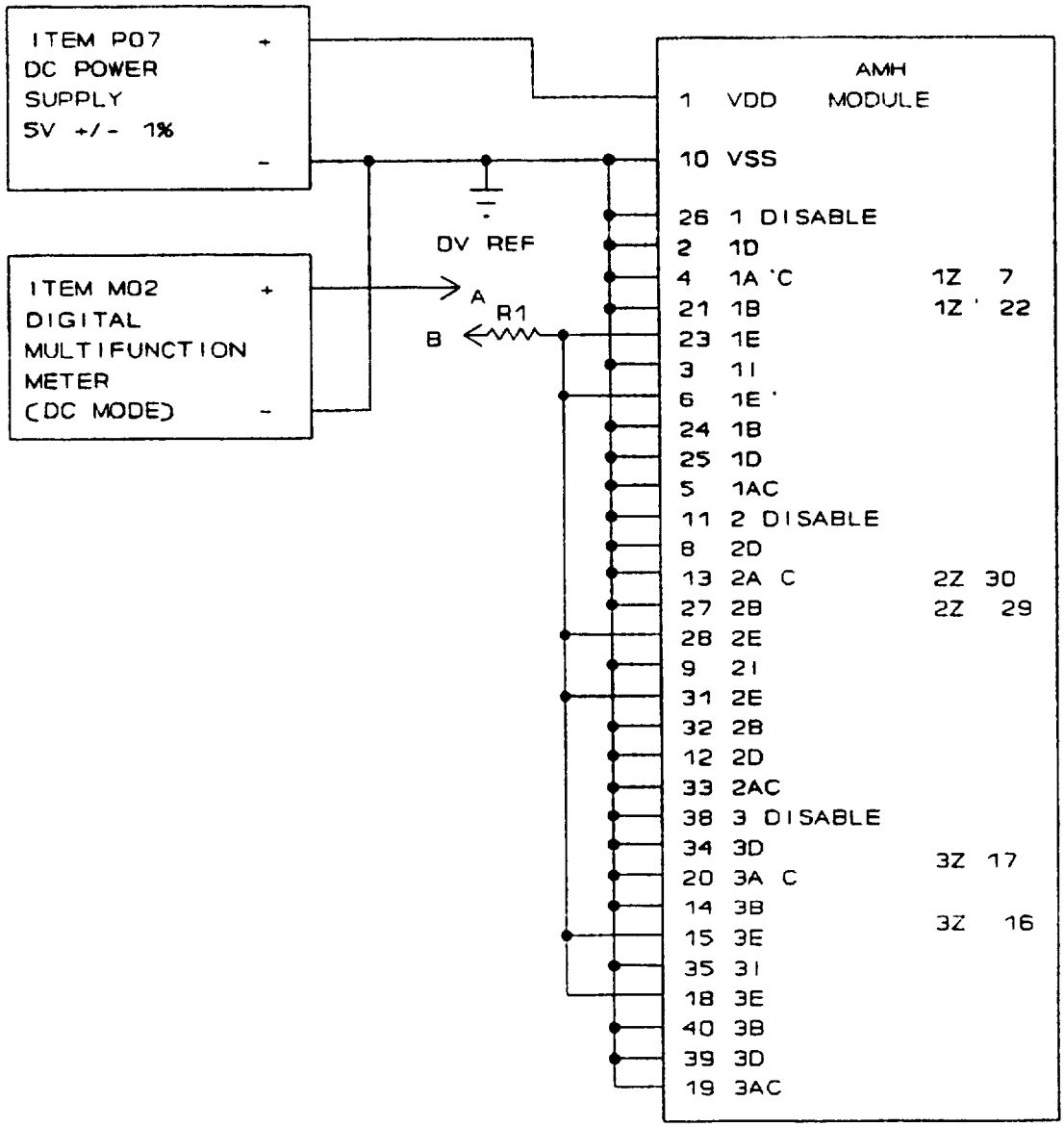
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NOTES:
1. **R1 = 9.09 kΩ ± 1%, 1/8 W.**
2. **R2 = 13.7 kΩ ± 1%, 1/8 W.**
3. **Since items P07-1 and P07-2 and items M02-1 and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.**

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

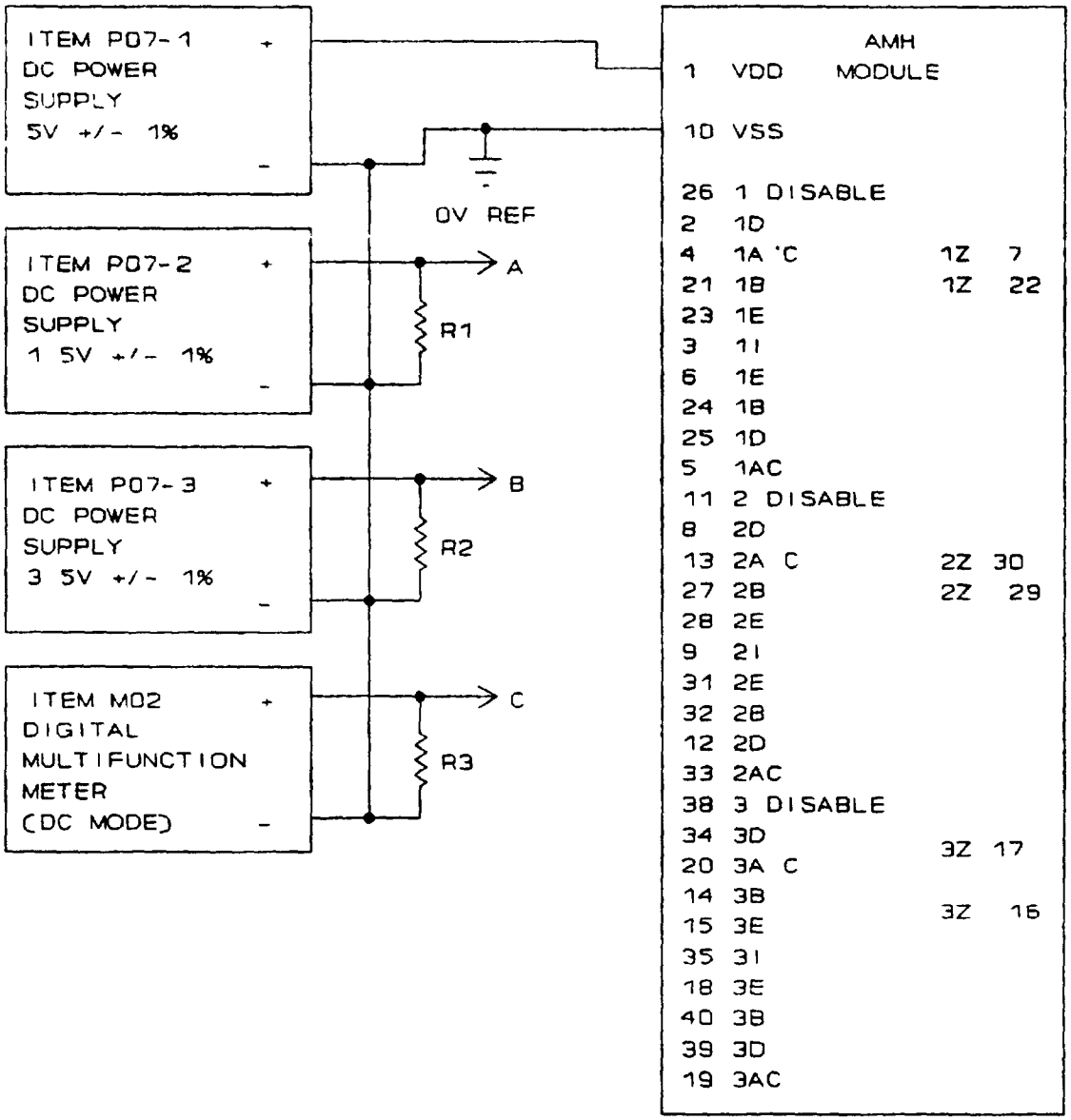
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NOTE: R1 = 1.5 kΩ ± 1%, 1/4 W.

FIGURE 8. Static output voltage (high-level and low-level) test circuit for tables VIII and IX.

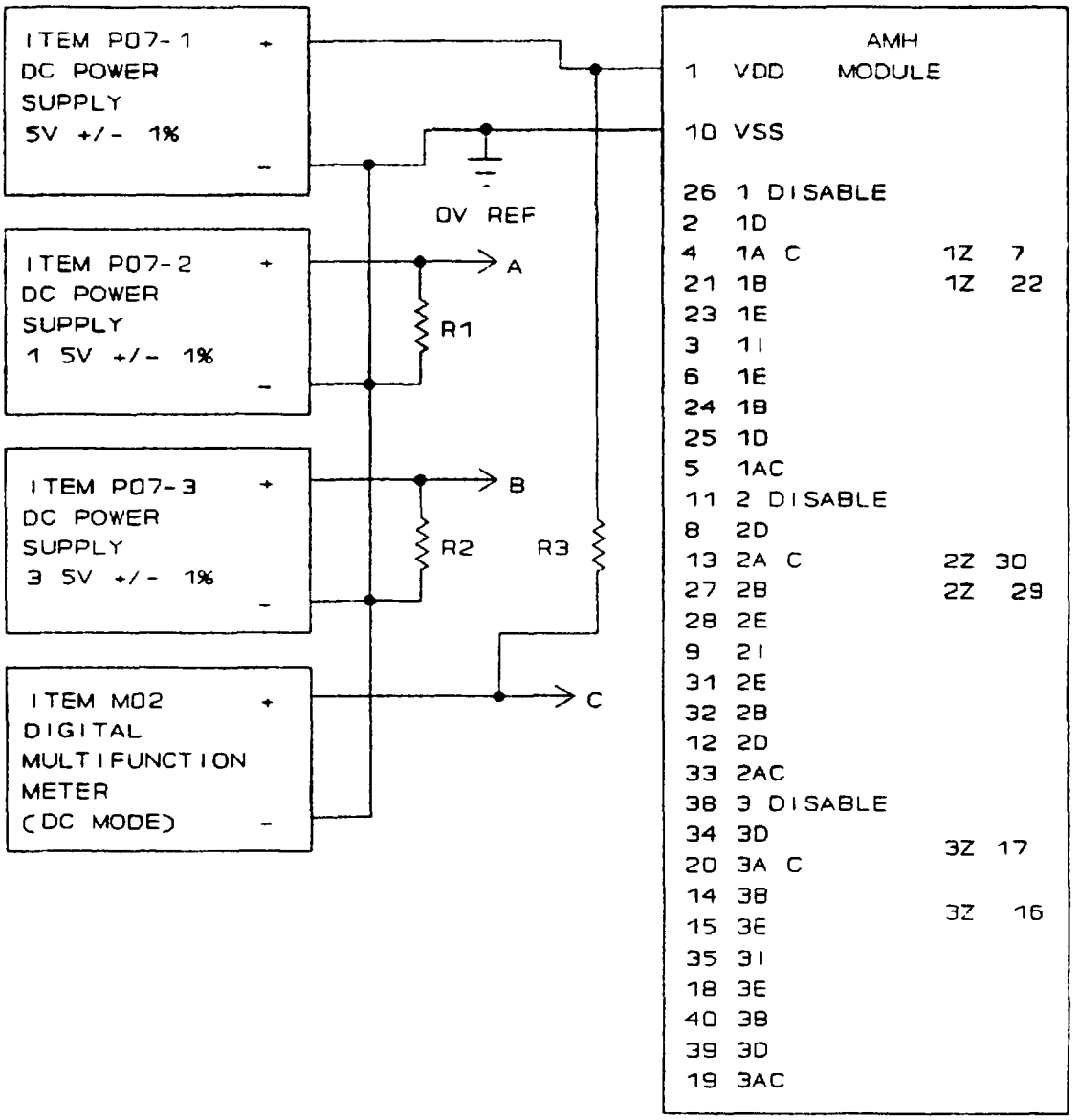
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NOTES:
1. R1 = R2 = 51.1 Ω ± 5%, 1/2 W.
2. R3 = 18.2 kΩ ± 1%, 1/8 W at -55°C.
   23.2 kΩ ± 1%, 1/8 W at 0° and 25°C.
   27.4 kΩ ± 1%, 1/8 W at 60°C.
   32.4 kΩ ± 1%, 1/8 W at 85°C.

FIGURE 9. Static output voltage (high-level) test for table X.

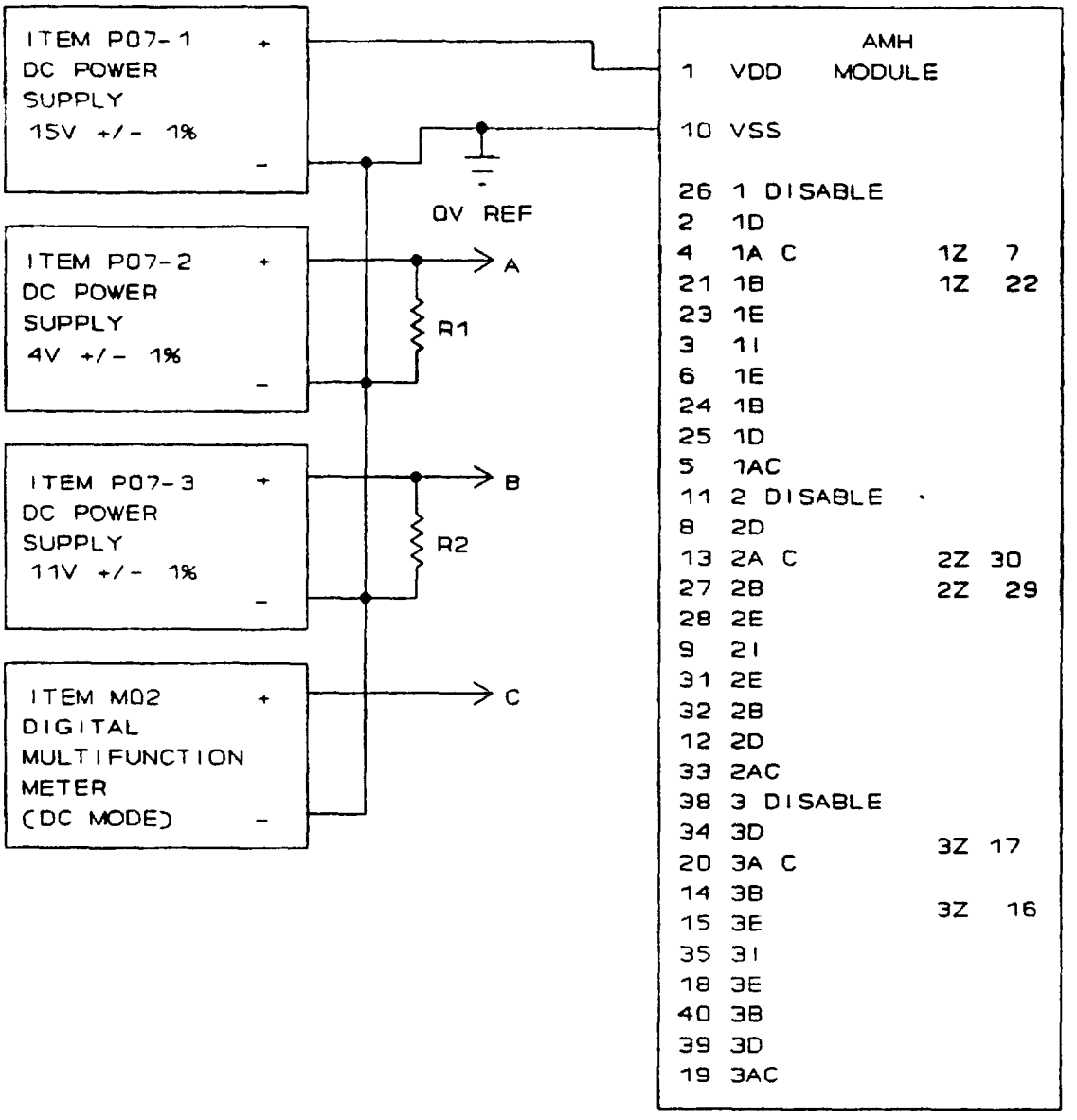
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NOTES:
1. R1 = R2 = 51.1 Ω ± 5%, 1/2 W.
2. R3 = 7.15 kΩ ± 1%, 1/8 W at -55°C.
   9.09 kΩ ± 1%, 1/8 W at 0° and 25°C.
   10.5 kΩ ± 1%, 1/8 W at 60°C.
   12.7 kΩ ± 1%, 1/8 W at 85°C.

FIGURE 10. Static output voltage (low-level) test for table XI.

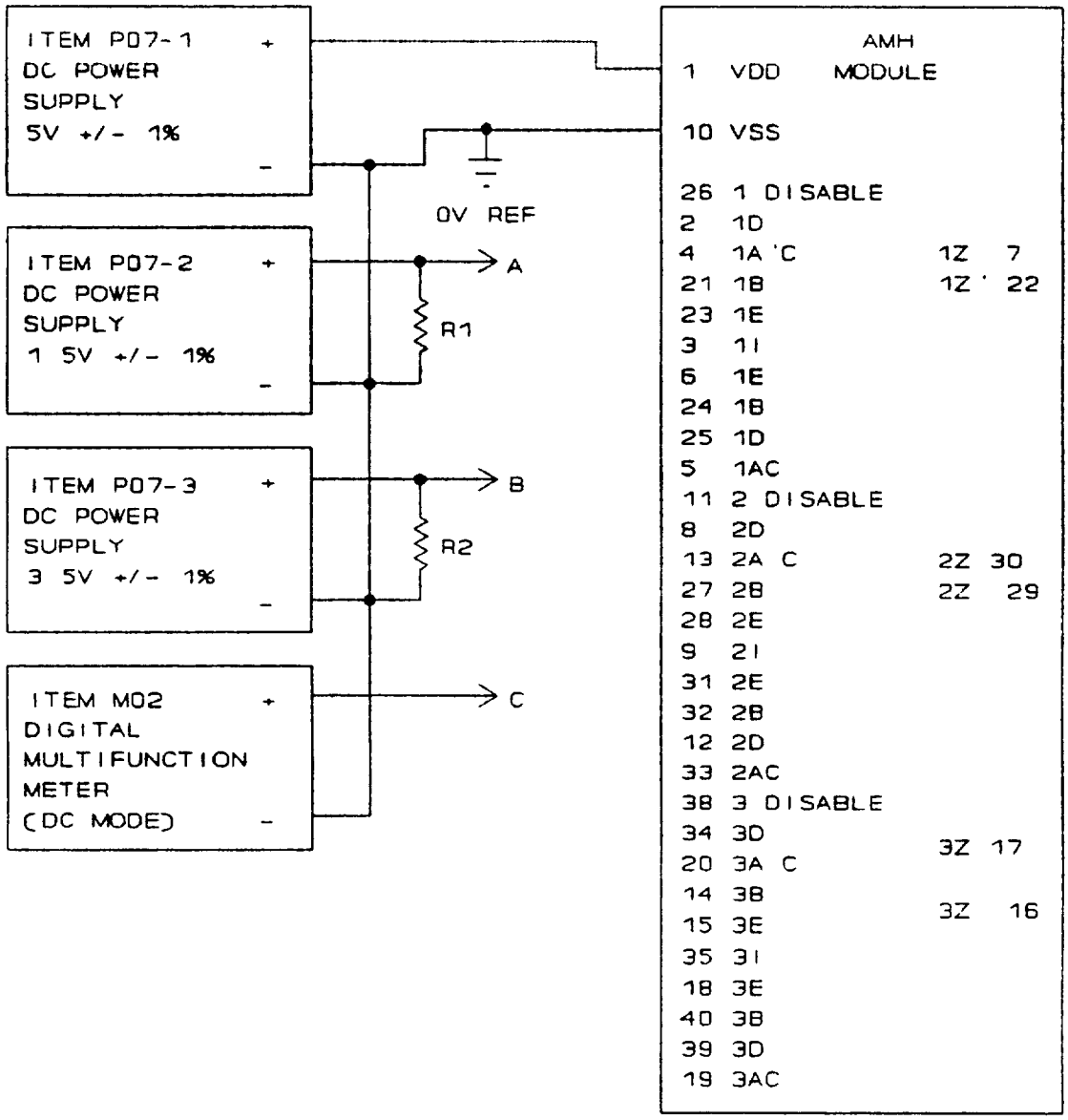
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NOTE: R1 = R2 = 51.1 Ω ± 5%, 5 W.

FIGURE 11. Static output voltage (high-level and low-level) test circuit for tables XII and XIII.

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NOTE: R1 = R2 = 51.1 Ω ± 5%, 1/2 W.

FIGURE 12. Functional test circuit for table XIV.

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